



# **Taming the Big Data Beast with Hadoop and Alternatives**

Turn the clock back to 1970—the year in which Edgar Codd published the paper ‘A Relational Model of Data for Large Shared Data Banks’. In it he described a language called ‘Structured Query Language’ (SQL), which went on to spearhead the RDBMS era. It became the language of choice for accessing RDBMSs. Forty-two years later, we are again at the brink of a data revolution—the Big Data revolution.

**T**he volume, velocity and variety of data (the so-called three Vs) are often cited as the characteristics of big data. Most of the producers of big data handle volumes exceeding 10 terabytes of data, some even exceeding 1 petabyte. In terms of velocity, or the rate at which the data is produced, 30 per cent of big data producers are bringing out more than 100 gigabytes of data every day. In terms of variety, while 55 per cent of producers are bringing out structured data, the rest are producing unstructured data. While SQL is ideal for processing structured data, it cannot handle unstructured data well. New data-processing solutions are needed to handle unstructured data. Hadoop addresses this need. It is a scalable solution for handling large volumes of unstructured data.

## **What is Hadoop?**

Hadoop has its origins in Google’s MapReduce—a computational paradigm introduced by Google to carry out distributed computations on large sets of data on a cluster of commodity hardware. The large data is itself stored in a distributed file system, so that it can be accessed as a contiguous volume in a transparent manner. The distributed file system also needs to provide redundancy, to prevent any data loss. MapReduce consists of two steps—a Map phase and a Reduce phase. In the Map phase, a master node partitions the input into sub-problems and distributes these tasks to worker nodes. The worker nodes process the assigned inputs and pass results back to the master. In the Reduce phase, the master node takes the answers collected from worker nodes for sub-problem tasks, and ‘reduces’